

## Git and GitHub Installation and Setup

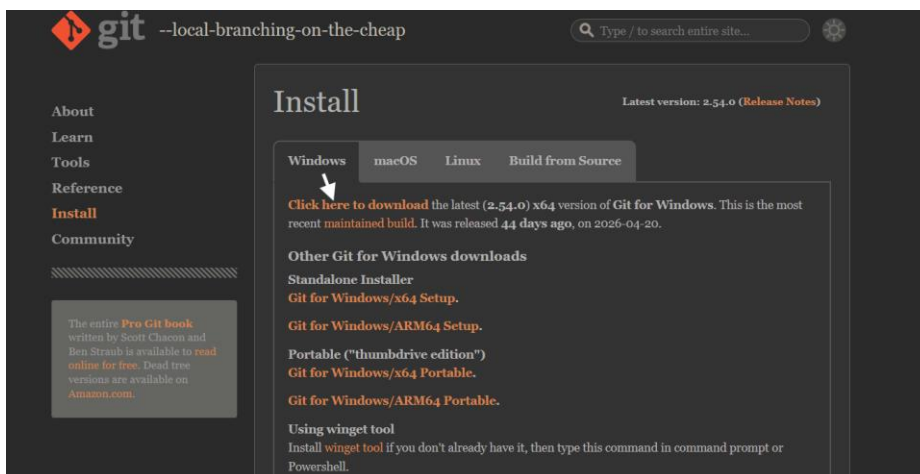
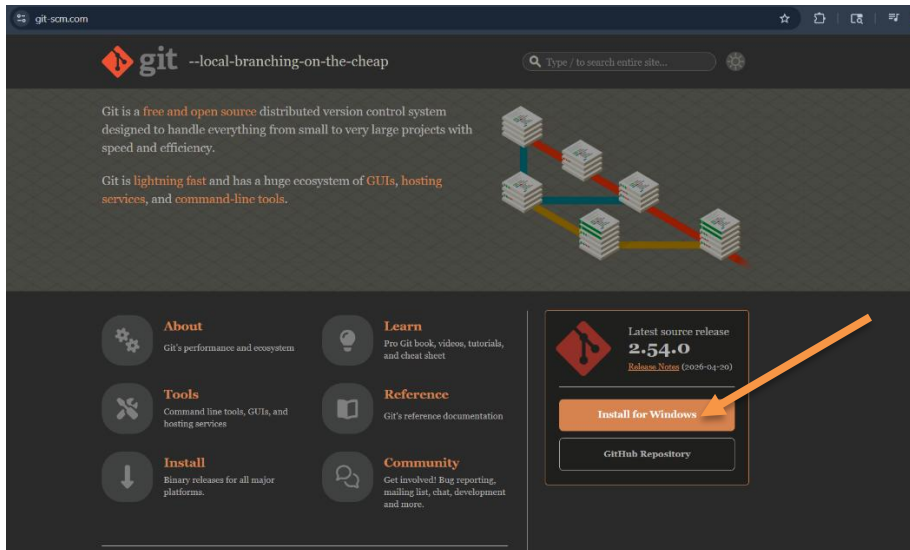
This guide is designed for students who have never used Git or GitHub before.

### 1. Create a GitHub Account

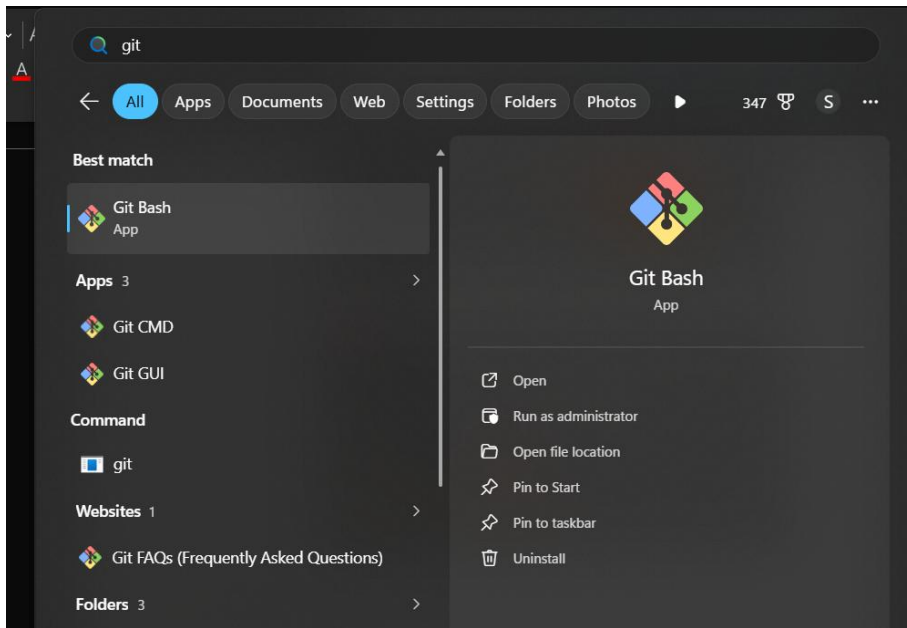
Visit <https://github.com> and create a free account. Verify your email address after registration.

### 2. Install Git

Download Git from <https://git-scm.com/downloads> and complete the installation using the default options.



Open Git Bash(Run following commands sequentially) and verify the installation:



> *git --version*

//this would give the git version hence verify its existence

>*pwd*

//will provide with the path

```
MINGW64:/c/Users/sgn07
sgn07@MINGW64 ~
$ git --version
git version 2.53.0.windows.1

sgn07@MINGW64 ~
$ pwd
/c/Users/sgn07

sgn07@MINGW64 ~
$ |
```

### 3. Configure Git (First Time Only)

Run these commands in Git Bash, Command Prompt, PowerShell, or the VS Code Terminal:

```
>git config --global user.name "YourGitHubUsername"
>git config --global user.email "your_email@example.com"
>git config --global init.defaultBranch main
```

Verify the configuration:

```
>git config --global --list
```

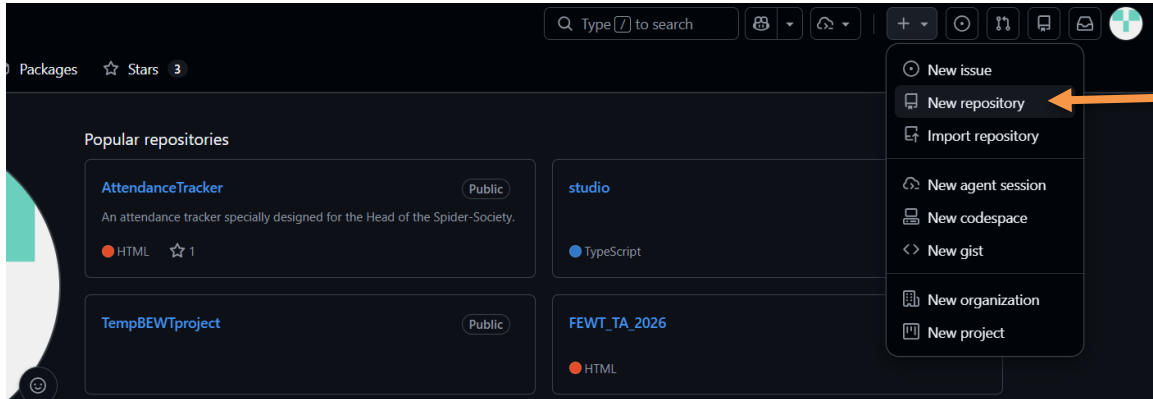
Why is this required?

- Git records your name and email with every commit.
- GitHub links commits to your account when the email matches your GitHub email.

#### 4. Create a Repository on GitHub

A repository (repo) is a project folder stored on GitHub.

1. Log in to GitHub.
2. Click New Repository.

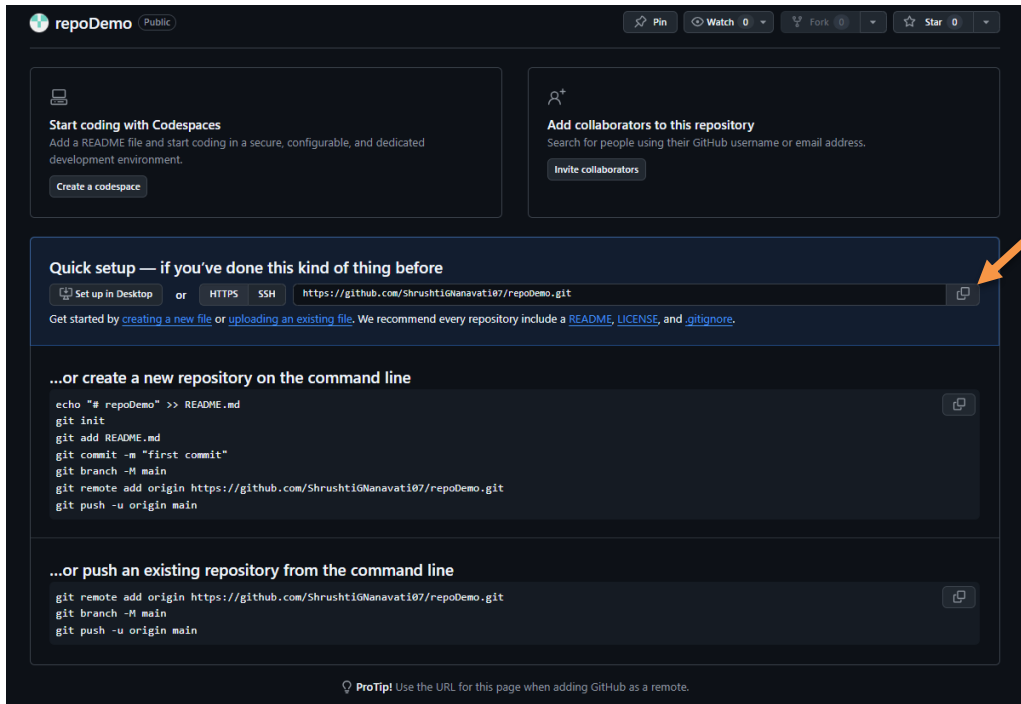


3. Enter a repository name.

A screenshot of the GitHub 'Create new repository' form. The form is divided into three sections: 1. General, 2. Configuration, and 3. Jumpstart your project with Copilot (optional). In the 'General' section, the 'Owner' is 'ShrushtiGNanavati07' and the 'Repository name' is 'repoDemo', with a green checkmark indicating it is available. The 'Description' field is empty. In the 'Configuration' section, 'Choose visibility' is set to 'Public', 'Add README' is 'Off', 'Add .gitignore' is 'No .gitignore', and 'Add license' is 'No license'. In the 'Jumpstart your project with Copilot' section, there is a text area for a 'Prompt' with the placeholder text 'Describe what you want Copilot to build'. A green 'Create repository' button is at the bottom right.

4. Click Create Repository.

5. Copy *REPOSITORY\_URL* from:



## 5. Recommended Folder Structure(Make this folder structure)

Sem3

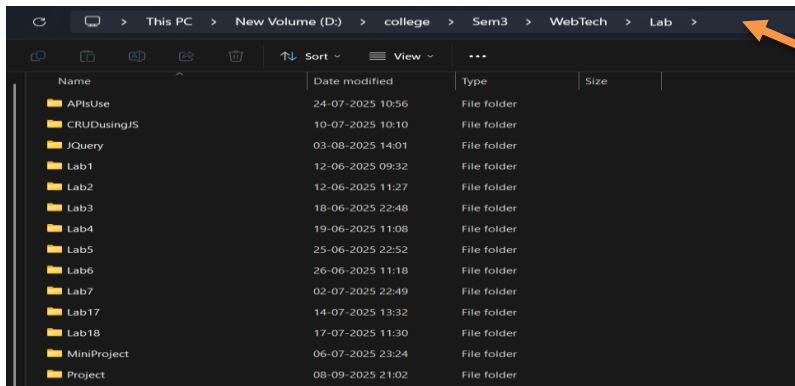
- └─ FEWT
  - └─ Labs
    - └─ Lab1

└─ a1.html (Save this file)

- └─ Lab2
- └─ Lab3

## 6. Upload Lab Files to GitHub

Open a terminal in the Labs folder and run:



Open command prompt here:

➤ *cmd*

*>git init*

(This creates D:/Sem3/FEWT/Lab/Lab1/.git)

*>git remote add origin* **REPOSITORY\_URL**

*>git remote -v*

(to verify)

Output: origin repo link (fetch)

*>git add Lab1*

(To add Lab1)

*>git status*

*>git commit -m "Added Lab1"*

*>git push -u origin main*

If your GitHub authentication is left this may open an authentication page. Kindly authenticate your account yourself and repeat the last step.

## 7. Upload Additional Labs

*> git add Lab2*

*> git commit -m "Added Lab2"*

*> git push*

## 8. Common Commands

Command

Purpose

git init

Create a Git repository

git add

Prepare files for commit

git commit

Save a snapshot of changes

git status

Show current repository status

git push

Upload changes to GitHub

## 9. Troubleshooting(Problems u may encounter)

1. 'git' is not recognized -> Reinstall Git and restart the terminal.
2. Authentication window does not open -> Update Git and sign in through your browser.
3. remote origin already exists -> Remove the old remote and add the correct one.
4. src refspec main does not match any -> Create a commit before pushing.